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Predicting delayed kidney graft function with gene expression in preimplantation biopsies and first-day posttransplant blood

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ABSTRACT

The purpose of this study was to investigate possible markers for predicting delayed graft function (DGF). To this end we analyzed, in pre-implantation biopsies (PIB) and in first-day post-Tx peripheral blood mononuclear cells (PBMC), the expression of five genes (*ACSL4*, *CUBN*, *DEFB1*, *FABP3*, *GK*) through real-time TaqMan PCR assays. These genes were selected from a large scale gene expression study in PIB. *DEFB1*, *FABP3* and *GK* expression levels in PIB were lower in cases with DGF and, in a multivariate analysis which included these genes and clinical variables, only *FABP3* expression remained independently associated with DGF. *FABP3* expression lower than −1.32 units of relative expression conferred an odds ratio for DGF of 41.1. Compared to the PBMC of recipients without DGF, recipients with prolonged DGF (pDGF) had lower *ACSL4* and higher *DEFB1* expression levels. In a multivariate analysis, including PBMC gene expression levels of *ACSL4*, *DEFB1* and *TLR4* (data from a previous study with the same patients) and clinical variables, only *TLR4* remained independently associated with pDGF. In summary, this study revealed *FABP3* expression in PIB as a marker for DGF and disclosed new genes possibly involved in the pathogenesis of DGF.

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1. Introduction

As the need of kidneys for transplantation remains higher than their availability from deceased donors (DDs), an increasing number of kidneys from expanded criteria, as well as from non-heart beating, donors are being used, leading to an increase in the incidence of delayed graft function (DGF) [1–4]. DGF is an important early complication of kidney transplantation that not only increases early morbidity, resulting in prolonged hospitalization and higher costs in the short-term, but is also associated with increased risk for acute rejection and for long-term graft failure.

The search for markers that could predict the occurrence of DGF has been the object of several studies [5–12], since they could serve to guide the implementation of pre- and posttransplant approaches for preventing or minimizing the damages of DGF [13]. A recent comprehensive review of the literature concluded that among a plethora of appraisal tools, only very few demonstrated predictive powers that could be considered in clinical practice [14].

Previous studies from our laboratory on DGF prediction investigated the expression levels, in preimplantation biopsies (PIB) and in first-day post-transplant peripheral blood leukocytes, of apoptosis and innate immunity related genes [9,11]. The major finding of these studies was that the concomitant increase in *BAX*/*BCL2* expression ratio in PIB and decrease in *TLR4* expression in peripheral blood leukocytes represents a very powerful predictor of DGF, with a positive prediction value (PPV) of 91% and a negative predictive value (NPV) of 92%, in reference to patients without any of these markers [11].

Abbreviations: ACSL4, acyl-CoA synthetase long-chain family member 4; AUC, area under the receiver operating characteristic curve; CUBN, cubilin; DD, deceased donor; *DEFB1*, defensin, beta 1; DGF, delayed graft function; *FABP3*, fatty acid binding protein 3; FFPE, formalin-fixed and paraffin-embedded; *GK*, glycerol kinase; PIB, preimplantation biopsies; PBMC, peripheral blood mononuclear cells; pDGF, prolonged delayed graft function; ROC, receiver operating characteristic; uRE, units of relative expression.

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Continuing our search of DGF related genes, the present study aimed to investigate, in PIB and in first-day post-transplant peripheral blood mononuclear cells (day 1-PBMC), the expression of candidate genes derived from a large scale expression study conducted in our laboratory (results deposited in the Gene Expression Omnibus (GEO), accession: GSE54888) that compared PIB from cases that evolved with and without DGF. For the current study, we performed a restrictive analysis on this data and we selected five genes, all located in different chromosomes, to be further evaluated as predictors of DGF: *ACSL4* (acyl-CoA synthetase long-chain family member 4), *CUBN* (cubilin), *DEFB1* (defensin, beta 1), *FABP3* (fatty acid binding protein, 3) and *GK* (glycerol kinase). The expression of these genes was evaluated in a different group of PIB and also in PBMC from the first-day posttransplant.

2. Materials and methods

2.1. Gene selection

Candidate genes evaluated in this study were selected from a large scale gene expression study performed in our laboratory which analyzed 54 PIB from DD (GEO accession no. GSE54888). Among the 54 PIB, 27 corresponded to transplants in recipients who presented DGF, of whom 13 had pDGF. Differentially expressed genes between groups with and without DGF, or between groups with pDGF and without DGF, were selected using random-variance Student's *t*-test ($p < 0.001$) with BRB ArrayTools version 4.2.1, developed by Dr. Richard Simon and BRB-ArrayTools Development Team (http://linus.nci.nih.gov/BRB-ArrayTools.html). In the comparisons between DGF vs. no DGF and between pDGF vs. no DGF, 50 and 131 differentially expressed genes were identified, respectively. Among these, we selected five genes (*ACSL4*, *CUBN*, *DEFB1*, *FABP3*, and *GK*) that had: (i) *p*-value <0.01 in both comparisons (DGF vs. no DGF and pDGF vs. no DGF); (ii) fold change $\geqslant 1.40$ or $\leqslant 0.71$ in at least one comparison; (iii) consistent direction of expression change in both comparisons (fold change >1 or <1) and (iv) official gene symbol.

2.2. Patients and samples

All transplants were performed in our center, between January 2008 and December 2010, and all recipients signed an informed consent form approved by the local ethics committee. Inclusion criteria were pretransplant negative complement-dependent cytotoxicity crossmatch against donor's T and B cells and no donor-specific HLA class I or class II antibodies with nMFIs > 1500 when assayed by Luminex single antigen bead assay (LabScreen Single Antigen, One Lambda, Part of Thermo Fisher Scientific, Canoga Park). All recipients had more than 18 years of age and received a kidney from a deceased donor. The immunosuppressive therapy consisted of a calcineurin inhibitor combined with prednisone and either azathioprine or mycophenolic acid, with or without induction with anti-thymocyte globulin or baziliximab.

The study on gene expression in preimplantation biopsy comprised 43 formalin-fixed and paraffin-embedded preimplantation kidney biopsies from DD transplants that were recovered from the histopathology archives of the Pathology Department, Escola Paulista de Medicina, Universidade Federal de São Paulo. Information on histological lesions of acute tubular necrosis was available from 35 biopsies.

The study on gene expression in day-1 PBMC included 73 transplant recipients.

DGF was defined as the need for dialysis within the first week after transplantation, except when only one session was performed immediately after surgery because of hyperkalemia or fluid overload. The decision to initiate dialysis originated from the transplant nephrologists, without any involvement from the study investigators. Prolonged DGF was defined as DGF lasting more than 14 days, with the end of DGF being considered as the day in which there was a decrease, in the patient off dialysis, of at least 20% in serum creatinine relative to the highest posttransplantation value.

Demographic and clinical information regarding the two groups of patients are shown in Table 1.

2.3. Sample processing, RNA extraction and reverse transcription

For RNA extraction from biopsies, four 20 μm sections were cut from the 43 archival blocks and placed in 1.5 mL RNase-free tubes. The sections underwent deparaffinization, proteinase digestion and the RNA was extracted with RecoverAll® Total Nucleic Acid Isolation kit (Life Technologies, Grand Island, NY), according to manufacturer's instructions. Total RNA (600 ng) was reverse transcribed in triplicates of 200 ng with SuperScript® VILO™ MasterMix (Life Technologies).

PBMC were isolated using Ficoll-Paque Plus (GE, Little Chalfont, BKM, UK) and underwent RNA extraction with the RNeasy® mini kit (QIAGEN, Valencia, CA), according to manufacturer's instructions. Total RNA (600 ng) was reverse transcribed in triplicates of 200 ng with RevertAid™ H Minus M-MuLV kit (Fermentas, Hanover, MD) and Oligo-dT primer (Integrated DNA Technologies, Coralville, IA).

2.4. Evaluation of gene expression levels

The real-time quantitative polymerase chain reaction (qPCR) was performed according to Applied Biosystems (Foster City, CA)

Table 1
Recipient, donor, and kidney characteristics in transplants with or without delayed graft function.

	Pre-implantation kidney graft biopsies (PIB) study		First-day post-transplant peripheral blood mononuclear cells study	
	DGF	No DGF	DGF	No DGF
Number of patients	21	22	47	26
Recipient age, years, mean ± standard deviation	49.0 ± 12.9	52.7 ± 11.5	48.6 ± 9.5	50.5 ± 12.7
Donor age, years, mean ± standard deviation	42.5 ± 11.8	42.6 ± 10.9	43.8 ± 14.0	43.2 ± 15.1
Cause of donor death, non-trauma event (*N*, %)	11 (52.4%)	18 (81.8%)	31 (67.4%)	13 (50.0%)
Extended criteria donor[a] (*N*, %)	8 (38.0%)	3 (13.6%)	13 (27.6%)	7 (26.9%)
Induction therapy with ATG or Basiliximab, (*N*, %)	17 (80.9%)	21 (95.4%)	30 (63.8%)	16 (61.5%)
Cold ischemia time, hours (median, min-max)	24.0 (16.3-33.8)	23.5 (12.0-30.0)	25.6 (11.1-37.2)	22.9 (8.3-37.0)
Moderate/severe acute tubular necrosis lesions in PIB (*N*, %)	12/15 (80.0%)	13/20 (65.1%)	19/25 (76.0%)	10/13 (76.9%)

DGF, delayed graft function; ATG, anti-thymocyte globulin.
Statistics: fisher's exact test for categorical variables; Mann–Whitney test for continuous variables.
No significant differences ($p < 0.05$) in any comparison between cases with and without DGF.
[a] According to United Network for Organ Sharing definition.

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protocols, using the following TaqMan Assay® reagents (Applied Biosystems): *ACSL4* (Hs00244871_m1), *CUBN* (Hs00153607_m1), *DEFB1* (Hs00608345_m1), *FABP3* (Hs00997360_m1), *GK* (Hs04235340_s1), *GUSB* (Hs00939627_m1), *HPRT1* (Hs02800695_m1), *POLR2K* (Hs01562397_m1), and *TBP* (Hs00427620_m1). All samples were run in triplicate on the StepOnePlus® sequence detection system and the results were analyzed with StepOne Software v2.2.2 (both from Applied Biosystems).

Relative quantification was performed by $2^{-\Delta\Delta Ct}$ method (Ct-threshold cycle) in which the expression of the target gene is normalized to an endogenous reference gene/s and to a calibrator (reference group) [15]. For normalization of gene expression in biopsies, we used the geometric mean of Ct values of three reference genes (*POLR2K*, *HPRT1*, *GUSB*) selected among eight genes cited in the literature as reference for gene expression in kidney tissue [16,17]. In our experience these three genes showed the most stable expression, according to the geNorm software version 3.4 (medgen.ugent.be/genorm) [18]. Gene expression levels in PBMC samples were normalized only by *TBP* expression levels, as in a previous study from our laboratory [9].

The reference groups consisted of samples from biopsies and from PBMC of cases that evolved without DGF. Expression levels were referred as units of relative expression (uRE).

2.5. Statistical analyses

Statistical analysis was performed with GraphPad Prism® 5.0, PASW Statistics, Version 18.0 (Chicago: SPSS Inc) and MedCalc®, Version 12.7.0. Categorical variables were analyzed using Fisher's exact test, whereas continuous variables were analyzed with Student's *t*-test. All gene expression values were $\log_2$ transformed and the outliers were defined by box-and-whiskers plots, according to Tukey [19]. Receiver operating characteristic (ROC) curves were used to determine cutoff values for gene expression levels as predictors of DGF. Stepwise logistic regression was used to evaluate the independence factors associated with DGF or pDGF. A p value <0.05 was considered significant.

3. Results

3.1. Donor, recipients and kidney characteristics

The baseline characteristics of donors and recipients included in the PIB study ($n = 43$) and in the day-1 PBMC study ($n = 73$) are shown in Table 1. Among the 43 PIB samples, 21 corresponded to recipients that presented DGF, of whom six had pDGF. Among the 73 PBMC samples, 47 were from recipients that presented DGF, of whom 20 had pDGF. There were no significant differences in recipient's or in donor's characteristics between cases with DGF, or pDGF, and without DGF, in any of the study groups. Acute tubular necrosis lesions were evaluated in 35 PIB and in 38 day-1 PBMC and no association of moderate/severe lesions with DGF was observed.

3.2. ACSL4, CUBN, DEFB1, FABP3 and GK expression levels in PIB according to occurrence of DGF

The expression levels of *DEFB1*, *FABP3* and *GK* were lower in cases with than in those without DGF, whereas no differences were observed concerning *ACSL4* and *CUBN* expression levels (Fig. 1). Next, we performed a multivariate regression analysis including the expression levels of *DEFB1*, *FABP3* and *GK*, cold ischemia time, donor age and the use of extended criteria donor. Only *FABP3* expression levels (OR = 6.96, 95% confidence interval [CI] = 01.92–25.29, $p = 0.003$) were independently associated with DGF occurrence.

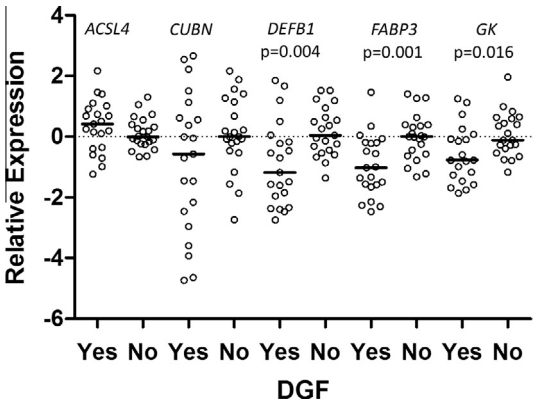

Fig. 1. *ACSL4*, *CUBN*, *DEFB1*, *FABP3* and *GK* expression levels in preimplantation biopsies from kidneys of deceased donors. Comparison between cases with ($n = 21$) and without ($n = 22$) DGF. Relative gene expression values were $\log_2$ transformed and the outlier values were not included in the analysis. Statistics: Student's *t*-test.

ROC curves were constructed to assess the potential of *FABP3* expression levels to predict DGF and the area under the curve (AUC) was 0.78 (95% CI, 0.64–0.92, $p = 0.002$). The cutoff that gave the best prediction for DGF was $\leqslant -1.32$ uRE and the odds ratio (OR), sensitivity, specificity, PPV, and NPV for DGF associated with low *FABP3* expression are presented in Table 2.

3.3. ACSL4 and DEFB1 expression levels in first-day posttransplant PBMC according to occurrence and duration of DGF

Since the selection of the genes to be tested in the present study derived from a study in PIB, before testing them in all PBMC samples, we checked their expression in a pool of 40 PBMC samples. As only *ACSL4* and *DEFB1* genes had detectable expression, only these two genes were tested in all individual PBMC samples. *ACSL4* and *DEFB1* expression levels did not differ between patients with and without DGF (Fig. 2A). However, lower *ACSL4* and higher *DEFB1* expression levels were observed in PBMC from recipients with pDGF as compared to those without DGF (Fig. 2B).

Since the group of patients used in this work was the same of our previous work showing the association of *TLR4* expression and pDGF [9] we included it in the multivariate regression analysis comprising *ACSL4* and *DEFB1* expression levels, cold ischemia time, donor age and extended criteria donor. The results showed that only *TLR4* expression (OR: 14.00; 95% CI:1.51–130.23, $p = 0.020$) remained independently associated with pDGF.

4. Discussion

This study aimed to investigate prediction of DGF based on gene expression in PIB and day-1 PBMC. The selection of the investigated genes, namely, *ACSL4*, *CUBN*, *DEFB1*, *FABP3*, and *GK*, derived from a previous large scale gene expression study from our laboratory, conducted in fresh PIB. In the current study, we used formalin-fixed and paraffin-embedded (FFPE), instead of fresh PIB, since this was the only PIB source from a different cohort of patients that was already available for testing. The use of FFPE PIB for gene expression studies represented a big challenge, since it is known that the RNA extracted from this type of material presents variable levels of degradation [20,21]. In fact, while the results of the present study confirmed that it is possible to use FFPE material for gene expression evaluation, they also highlighted the fact that a significant portion of the samples, in our case 28%, cannot be analyzed due insufficient amount or quality of RNA.

Table 2
Prediction of delayed graft function conferred by low *FABP3* expression levels in pre-implantation biopsies from kidneys of deceased donors.

uRE	DGF	No DGF	OR (95% CI)	p[a]	Sensitivity	Specificity	PPV	NPV	ACC
FABP3 $\leqslant -1.32$	10/21 (47.6%)	0/22 (0.0%)	41.1 (2.2–765.9)	0.0002	48%	100%	100%	67%	74%

uRE, units of relative expression $\log_2$ transformed; DGF, delayed graft function; OR, odds ratio; CI, confidence interval; PPV, positive predictive value; NPV, negative predictive value; ACC, accuracy.
[a] Fisher's exact test.

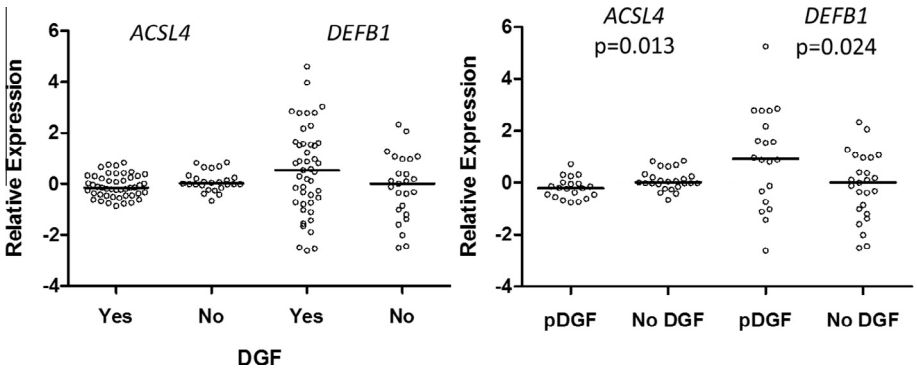

Fig. 2. *ACSL4* and *DEFB1* expression levels in first-day posttransplant peripheral blood mononuclear cells. (A) Comparison between cases with ($n = 47$) and without ($n = 26$) DGF. (B) Comparison between cases with prolonged DGF (pDGF) ($n = 20$) and without DGF ($n = 26$). Relative gene expression values were $\log_2$ transformed and the outlier values were not included in the analysis. Statistics: Student's *t*-test.

The expression of *DEFB1*, *FABP3*, and *GK* was lower in the biopsies of cases with DGF than in those without, whereas no statistically significant differences were observed concerning the other two genes (*ACSL4* and *CUBN*). An explanation for the association of low *GK* expression with DGF could be that lower levels of the encoded enzyme, glycerol kinase, would compromise the phosphorylation of glycerol. This would result in increased glycerol levels, which was shown to be nephrotoxic and able to induce acute kidney injure in mice [22,23]. On the other hand, the lower levels of *FABP3* in the DGF group could be interpreted as a decrease in the protection against ischemic damage, considering that fatty acid-binding proteins were shown to minimize the tissue damage caused by ischemia by reducing the effects of oxidative stress through the binding of fatty acids and products of lipid peroxidation [24–26]. Finally, the association of lower expression levels of *DEFB1* with DGF, could simply be reflecting a decrease in the number of tubular cells due to ischemia-induced cell death by necrosis and apoptosis, since this gene encodes an antimicrobial peptide of urogenital tissues constitutively produced by renal tubular cells [27–29].

In a multivariate analysis, including the three genes differentially expressed and cold ischemia time, donor age and extended criteria donor, only *FABP3* expression levels remained significant. With respect to the ability of low *FABP3* expression levels to predict DGF, considering the cutoff $\leqslant -1.32$ uRE suggested by the ROC analysis, although the sensitivity was only 48%, the specificity was 100%. In the future, we plan to evaluate, in PIB, the expression of *FABP3* in conjunction with the expression of *BAX* and *BCL2* genes, since in a previous study we have found that a high *BAX*/*BCL2* expression ratio in PIB was associated, contrary to *FABP3* expression, with a high sensitivity (80%) but with a low specificity (56%) [11].

In addition to the PIB study, we sought to analyze, in another group of patients, the relationship between gene expression levels in day 1-PBMC and DGF. *GK*, *FABP3*, and *CUBN* expression levels were not analyzed because a preliminary analysis failed to detect expression of these genes in PBMC. Concerning *ACSL4* and *DEFB1* expression in day 1-PBMC, no differences were observed in the comparison of cases with and without DGF, but we identified an association between the expression levels of these genes and pDGF, being *ACSL4* levels lower, and *DEFB1* levels higher, in cases with pDGF, when compared to those without DGF. In the PIB study, it was not possible to analyze the association between gene expression levels and pDGF due to the small number of patients with pDGF in this study group. The protein encoded by the *ACSL4* gene converts free long-chain fatty acids into fatty acyl-CoA esters, thereby playing a key role in lipid biosynthesis and fatty acid degradation [30]. However, we do not have any hypothesis to explain the relationship between this effect and pDGF. Concerning *DEFB1* expression it is interesting to note that this beta-defensin, constitutively produced by renal tubular cells, shows increased expression in blood monocytes and macrophages upon activation with lipopolysaccharide and/or interferon-gamma [31,32]. Thus, whereas the association of DGF with decreased *DEFB1* expression in PIB could be related to loss of renal tubular cells, the increased expression of *DEFB1* in PBMC in cases of pDGF could indicate a sustained inflammation contributing to the prolongation of DGF.

In a previous investigation, conducted with the same first-day blood samples used in the present study, we found an association of low *TLR4* expression with both DGF and pDGF [9]. Therefore, we tested whether the combination of low *TLR4*, low *ACSL4* and high *DEFB1* expression could further improve the prediction of pDGF, but in a multivariate analysis, only low *TLR4* expression levels remained independently associated with pDGF. This result does not mean, however, that we should discard a role of *ACSL4* or *DEFB1* encoded molecules in the pathogenesis of pDGF.

In conclusion, this study revealed low *FABP3* expression in PIB as a new marker for DGF and disclosed new genes, whose expression in PIB or in day 1-PBMC might be involved in the pathogenesis of DGF or pDGF.

5. Disclosure

The authors declare no conflicts of interest.

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References

[1] M.J. Englesbe, Y. Ads, J.A. Cohn, C.J. Sonnenday, R. Lynch, R.S. Sung, et al., The effects of donor and recipient practices on transplant center finances, Am. J. Transplant. 8 (2008) 586–592.
[2] S.G. Yarlagadda, S.G. Coca, R.N. Formica, E.D. Poggio, C.R. Parikh, Association between delayed graft function and allograft and patient survival: a systematic review and meta-analysis, Nephrol. Dial. Transplant. 24 (2009) 1039–1047.
[3] D.M. Summers, R.J. Johnson, J. Allen, S.V. Fuggle, D. Collett, C.J. Watson, et al., Analysis of factors that affect outcome after transplantation of kidneys donated after cardiac death in the UK: a cohort study, Lancet 376 (2010) 1303–1311.
[4] N.M. Butala, P.P. Reese, M.D. Doshi, C.R. Parikh, Is delayed graft function causally associated with long-term outcomes after kidney transplantation? Instrumental variable analysis, Transplantation 95 (2013) 1008–1014.
[5] L.K. Kayler, R. Mohanka, A. Basu, R. Shapiro, P.S. Randhawa, Correlation of histologic findings on preimplant biopsy with kidney graft survival, Transpl. Int. 21 (2008) 892–898.
[6] W.D. Irish, J.N. Ilsley, M.A. Schnitzler, S. Feng, D.C. Brennan, A risk prediction model for delayed graft function in the current era of deceased donor renal transplantation, Am. J. Transplant. 10 (2010) 2279–2286.
[7] K. Kotsch, K. Kunert, V. Merk, A. Reutzel-Selke, A. Pascher, F. Fritzsche, et al., Novel markers in zero-hour kidney biopsies indicate graft quality and clinical outcome, Transplantation 90 (2010) 958–965.
[8] R. Korbély, J. Wilflingseder, P. Perco, A. Kainz, R.M. Langer, B. Mayer, et al., Molecular biomarker candidates of acute kidney injury in zero-hour renal transplant needle biopsies, Transpl. Int. 24 (2011) 143–149.
[9] V. Andrade-Oliveira, E.F. Campos, A. Goncalves-Primo, P.C. Grenzi, J.O. Medina-Pestana, H. Tedesco-Silva, et al., TLR4 mRNA levels as tools to estimate risk for early posttransplantation kidney graft dysfunction, Transplantation 94 (2012) 589–595.
[10] C. Kreepala, K.S. Famulski, J. Chang, P.F. Halloran, Comparing molecular assessment of implantation biopsies with histologic and demographic risk assessment, Am. J. Transplant. 13 (2013) 415–426.
[11] A. Goncalves-Primo, T.B. Mourão, V. Andrade-Oliveira, E.F. Campos, J.O. Medina-Pestana, H. Tedesco-Silva, et al., Investigation of apoptosis-related gene expression levels in preimplantation biopsies as predictors of delayed kidney graft function, Transplantation 97 (2014) 1260–1265.
[12] I. Fonseca, H. Reguengo, M. Almeida, L. Dias, L.S. Martins, S. Pedroso, et al., Oxidative stress in kidney transplantation: malondialdehyde is an early predictive marker of graft dysfunction, Transplantation 97 (2014) 1058–1065.
[13] S.K. Benaragama, T. Tymkewycz, B.J. John, A. Davenport, B. Lindsey, D. Nicol, J. Olsburgh, et al., Do we need a different organ allocation system for kidney transplants using donors after circulatory death?, BMC Nephrol 15 (2014) 83–89.
[14] A.J. Dare, G.J. Pettigrew, K. Saeb-Parsy, Preoperative assessment of the deceased-donor kidney: from macroscopic appearance to molecular biomarkers, Transplantation 97 (2014) 797–807.
[15] K.J. Livak, T.D. Schmittgen, Analysis of relative gene expression data using real-time quantitative PCR and the 2(-delta delta C(T)), Methods 25 (2001) 402–408.
[16] J. Vandesompele, K. De Preter, F. Pattyn, B. Poppe, N. Van Roy, A. De Paepe, et al., Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes, Genome Biol. 3 (7) (2002). Research0034.1–0034.11.
[17] S.T. Glenn, K.L. Head, B.T. Teh, K.W. Gross, H.L. Kim, Maximizing RNA yield from archival renal tumors and optimizing gene expression analysis, J. Biomol. Screen. 15 (1) (2010) 80–85.
[18] K. Kashofer, C. Viertler, M. Pichler, K. Zatloukal, Quality control of RNA preservation and extraction from paraffin-embedded tissue: implications for rt-PCR and microarray analysis, PLoS One 8 (7) (2013) e70714.
[19] J.W. Tukey, Exploratory Data Analysis, Addison-Wesley Publishing Company, Reading, MA, 1977.
[20] C.D. Cohen, H.J. Gröne, E.F. Gröne, P.J. Nelson, D. Schlöndorff, M. Kretzler, Laser microdissection and gene expression analysis on formaldehyde-fixed archival tissue, Kidney Int. 61 (2002) 125–132.
[21] S.T. Glenn, C.A. Jones, P. Liang, D. Kaushik, K.W. Gross, H.L. Kim, Expression profiling of archival renal tumors by quantitative PCR to validate prognostic markers, Biotechniques 43 (2007). 639–40, 642–3, 647.
[22] E. Homsi, P. Janino, J.B.L. de Faria, Role of caspases on cell death, inflammation, and cell cycle in glycerol-induced acute renal failure, Kidney Int. 69 (2006) 1385–1392.
[23] L. Rahib, N.K. MacLennan, S. Horvath, J.C. Liao, K.M. Dipple, Glycerol kinase deficiency alters expression of genes involved in lipid metabolism, carbohydrate metabolism, and insulin signaling, Eur. J. Hum. Genet. 15 (2007) 646–657.
[24] T. Nakata, A. Hashimoto, M. Hase, K. Tsuchihashi, K. Shimamoto, Human heart-type fatty acid-binding protein as an early diagnostic and prognostic marker in acute coronary syndrome, Cardiology 99 (2003) 96–104.
[25] A. Kamijo-Ikemori, T. Sugaya, A. Obama, J. Hiroi, H. Miura, M. Watanabe, et al., Liver-type fatty acid binding protein attenuates renal injury induced by unilateral ureteral obstruction, Am. J. Pathol. 169 (2006) 1107–1117.
[26] T. Yamamoto, E. Noiri, Y. Ono, K. Doi, K. Negishi, A. Kamijo, et al., Renal L-type fatty acid–binding protein in acute ischemic injury, J. Am. Soc. Nephrol. 18 (2007) 2894–2902.
[27] C. Zhao, I. Wang, R.I. Lehrer, Widespread expression of beta-defensin hBD-1 in human secretory glands and epithelial cells, FEBS Lett. 396 (2–3) (1996) 319–322.
[28] E.V. Valore, C.H. Park, A.J. Quayle, K.R. Wiles, P.B. McCray Jr, T. Ganz, Human beta-defensin-1: an antimicrobial peptide of urogenital tissues, J. Clin. Invest. 101 (1998) 1633–1642.
[29] R. Oberbauer, M. Rohrmoser, H. Regele, F. Mühlbacher, G. Mayer, Apoptosis of tubular epithelial cells in donor kidney biopsies predicts early renal allograft function, J. Am. Soc. Nephrol. 10 (1999) 2006–2013.
[30] E. Soupene, F.A. Kuypers, Mammalian long-chain acyl-CoA synthetases, Exp. Biol. Med. 233 (2008) 507–521.
[31] D. Yang, O. Chertov, S.N. Bykovskaia, Q. Chen, M.J. Buffo, J. Shogan, et al., Beta-defensins: linking innate and adaptive immunity through dendritic and T cell CCR6, Science 286 (5439) (1999) 525–528.
[32] L.A. Duits, B. Ravensbergen, M. Rademaker, P.S. Hiemstra, P.H. Nibbering, Expression of beta-defensin 1 and 2 mRNA by human monocytes, macrophages and dendritic cells, Immunology 106 (2002) 517–525.